

Compiler Construction

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COURSE OBJECTIVE:

- Provide an understanding of the fundamental principles in Compiler Construction.
- Learn the process of mapping and design a self-developed language to implement different parsing techniques in order to make efficient parsers.
- Provide the skills needed for building compilers for various situations that one may encounter in a career in Computer Science.

COURSE OUTLINE:

- Introduction to Interpreter and compilers , phases of compiler.
- Compiler Techniques and Methodology.
- Organization of Compilers.
 - Lexical and Syntax Analysis.
 - Parsing techniques, Types of parsers, Type checking, Semantic analyzer, Object code generation and optimization, detection and recovery from errors. Contrast between compilers and interpreter.

Course learning objectives

CLO #	Course Learning Outcomes (CLOs)	PLO Mapping	Bloom's Taxonomy
1	Discuss the fundamental concepts of the basic structure of compiler and recall R.E and FA.	PLO_1 (Academic Education)	C2 (Understanding)
2	Explain different methods and techniques through which one can design its own Language.	PLO_2 (Knowledge for Computing Problem)	C2 (Understanding)
3	Apply the Different parsers and draw connections of their own generated Grammar in any host Language	PLO_3 (Problem Analysis)	C3 (Applying)

Assessment tool

Assessment Tools	CLO1	CLO2	CLO3
Quizzes	3 (9.6%)	3 (8.3%)	4 (12%)
Assignments	3 (9.6%)	3 (8.3%)	4 (12%)
Mid	15(48.3%)	15(41.6)	-----
Final	10(32.2%)	15(41.6%)	25(76%)
Total	31 (31%)	36 (26%)	33 (33%)

Grading Policy

- **Assessment Tools:**

- Quizzes → 10 marks
- Assignments → 10 marks
- Midterm Exam → 30 marks
- Final Exam → 50 marks

- **Recommended Book:**

Aho, A. *et al.* (2015) ***Compilers: Principles, techniques, and Tools***. India: Pearson India Education Services.

- **Reference Books:**

Holub, A.I. (2006) *Compiler design in C*. New Delhi: Prentice-Hall of India.

- Keith D. Cooper , Linda Torczon (2022) *Engineering a Compiler*. Elsevier Science